

Aphasias and Common Stroke Syndromes

Adena Leder, DO FAAN

Associate Professor, Department
of OMM

aleder@nyit.edu

Office of Academic Affairs

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Session Objectives

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- Learning Objective 1-Identify the different types of aphasia.
- Learning Objective 2- Localize different stroke syndromes and the arteries that supply them.
- Learning Objective 3- Understand the difference between an arterial and a venous stroke.
- Learning Objective 4- Recognize the risk factors for venous infarcts.
- Learning Objective 5- Recognize the different types of intracranial hemorrhages.

Epidemiology of Stroke

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- Stroke is the 2nd leading cause of both death and disability worldwide.
- 87% of strokes are ischemic.

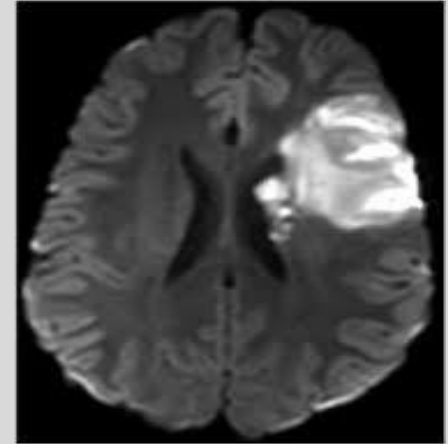


Image:itnonline.com

Neurology.Global Epidemiology of Stroke and Access to Acute Ischemic Stroke Interventions. Nov.16, 2021. Issue 97

Known Stroke Risk Factors

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Modifiable Risk Factors

- Tobacco use/Regular cannabis use
- Hypertension >130/80
- Dyslipidemia
- Diabetes
- Kidney disease
- Sleep apnea
- Heavy Etoh
- Diet
- Atrial Fibrillation

Non-modifiable Risk factors

- Male sex
- Age
- Genetic susceptibility
- Race



Source: Continuum Lifelong Learning in Neurology. Cerebrovascular Disease. Volume 26, Number 2. April 2020. Image: DW.COM

Mediterranean Diet

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- Associated with a lower stroke risk
- High intake of
 - Olive oil
 - Fruits and vegetables
 - Nuts and whole grains
 - Moderate intake fish and poultry



Case 1

- 72-year-old man with HTN, DM, HLD
- Right sided weakness
- Speech difficulty
- Follows commands, understands, reads
- Difficulty producing speech, unable to repeat or to write.

<https://youtu.be/RMa9BVpJkYQ>



Questions to Consider

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Which side of the brain is language function generally localized to?

Why do neurologists always ask if a patient is right or left handed?

Aphasia

- Disorder of language is based on:
 - a)spontaneous speech
 - b)comprehension
 - c)naming
 - d)repetition
 - e)writing
 - f)reading
- Often these disorders are not seen in isolation.

Types of Aphasia

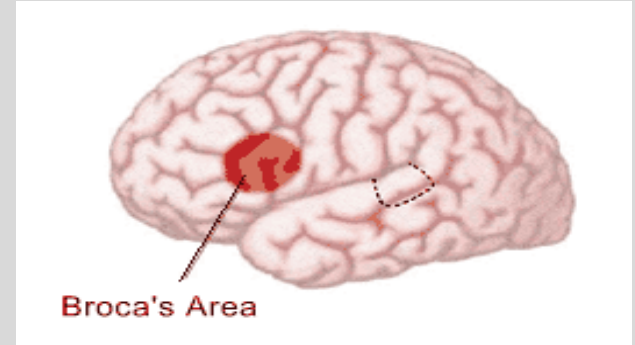
- Broca's aphasia (expressive)
- Wernicke's aphasia (receptive)
- Transcortical motor aphasia
- Transcortical sensory aphasia
- Global aphasia
- Conduction aphasia

Expressive Aphasia (Broca's)

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- Most commonly occurs as a result of very large left middle cerebral artery distribution infarcts. Extends involves the **frontal** lobe and often extends into the **superior temporal** lobe.
- Short simple sentences or one-two word utterances.
- Non-fluent

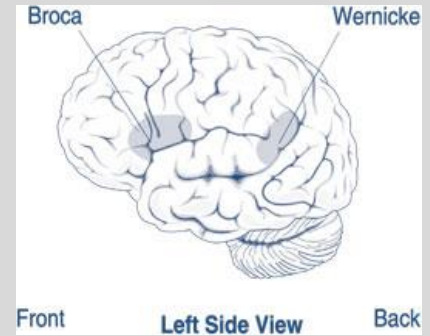


Receptive aphasia (Wernicke's)

- Stems from large middle cerebral artery infarcts.
- Lesion to left posterior perisylvian cortex.
- Poor speech comprehension (word deafness).
- Language is empty with lack of meaningful content.

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<https://youtu.be/3oef68YabD0>

Continuum:Lifelong Learning in Neurology. Behavioral Neurology and Psychiatry. Vol. 27, No.6. December 2021,

Transcortical Motor Aphasia

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- Stroke to left hemisphere, but spares the perisylvian cortex. Frontal lobe lesion.
- Nonfluent aphasia (hesitant speech)
- **Repetition intact**
- Good comprehension
- Anterior cerebral artery territory lesion.

Transcortical Sensory Aphasia

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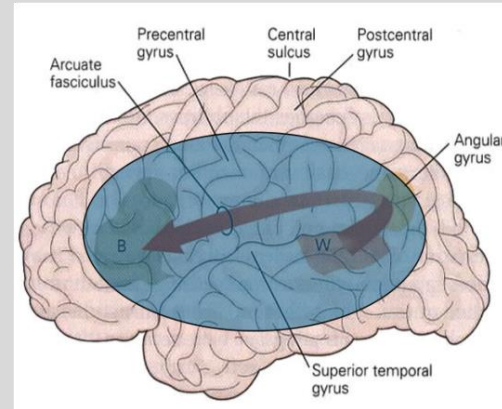
- Analogue of Wernicke's receptive aphasia
- Poor comprehension
- **Intact repetition, may be echolalic**
- Sparing of perisylvian cortex
- Reading and writing is impaired

Global Aphasia

- Expressive and receptive aphasia
- Speech is non fluent/mute
- Comprehension is poor
- Naming, repetition, reading and writing are affected.
- Inferior frontal and superior temporal gyrus lesion (L MCA)

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A lateral view of the human brain with the superior temporal gyrus highlighted in red. The label 'superior temporal gyrus' is written in red text across the highlighted area.

- [illegible]

Case 2

- 62-year-old man
- “I can’t see out of my left eye”
- How do you decide if this is the left eye or the left visual field?

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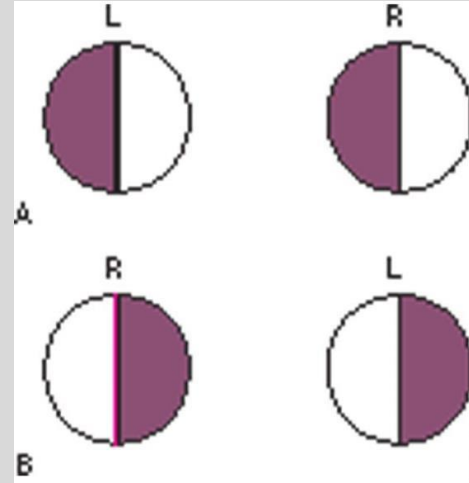
Posterior Cerebral Art Infarct

- PCAs are major terminal branches of the basilar artery. They supply branches to the midbrain, thalamus, occipital lobes and medial and inferior temporal lobes.
- Unilateral infarctions of occipital lobe may cause contralateral homonymous hemianopia

Uptodate.com December, 2024

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Neglect

- Inability to process information on one side of the body.
- More severe after right sided cerebral lesions. May occur after **right posterior parietal stroke**.
- May treat a body part as foreign



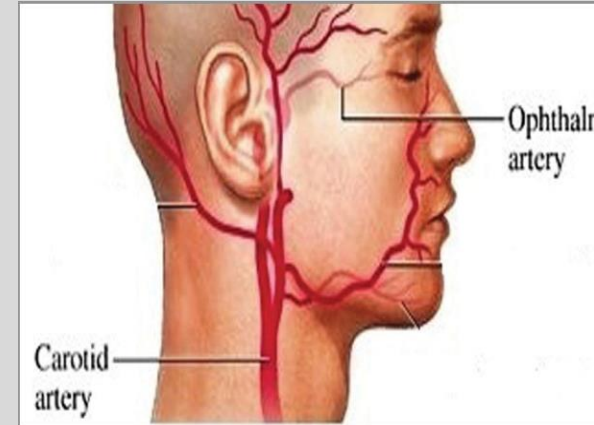
Amaurosis Fugax

- Transient monocular or binocular vision loss.
- Vascular cause for the vision loss. Shade being pulled down or constriction of the visual field.
- May distinguish ICA- a branch (central retinal artery occlusion) from MCA.

www.ipoecho.com. Uptodate.com 1/7/25

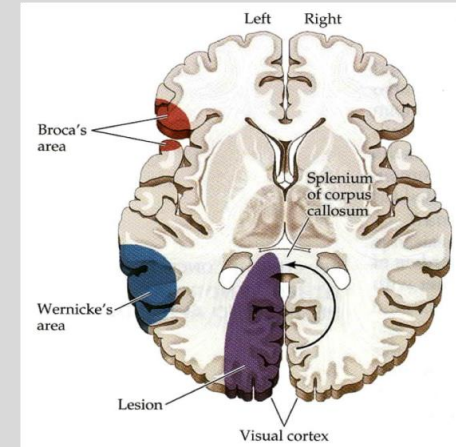
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Alexia without Agraphia

- Left posterior cerebral artery infarction.
- Left occipital region and splenium of the corpus callosum.
- Pure word blindness
- Unable to read, but able to write.



Gerstmann's Syndrome

- May be from PCA territory infarction affecting angular gyrus or from MCA involving angular gyrus.

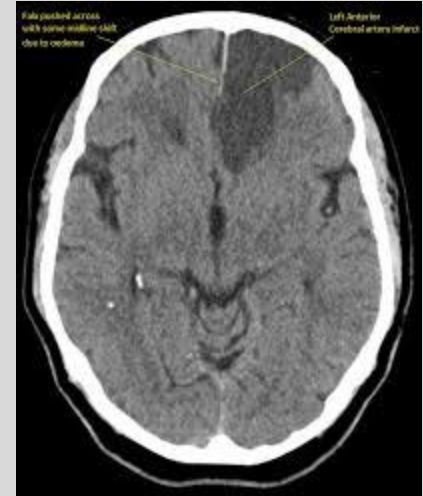


Anterior Cerebral Art. Infarct

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- Uncommon in isolation (<3%)
- ASA stroke can occur due to 2 main types of interruption to the brain's blood flow. 1) Blockage or clot 2) Hemorrhagic stroke due to aneurysm rupture. **Contralateral lower extremity> arm weakness.**
- Abulia, akinetic mutism impaired memory and emotional disturbances.



Uptodate.com 2024

Image; Neurovascularmedicine.com

Internal Carotid Artery Stroke

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- ACA and MCA are branches
- Atherosclerotic plaque/ critical stenosis at cervical bifurcation
- Contralateral hemiparesis and hemisensory loss (face>arm>leg) and motor>sensory
- Transient central retinal artery occlusion- better prognosis than hemispheric stroke.

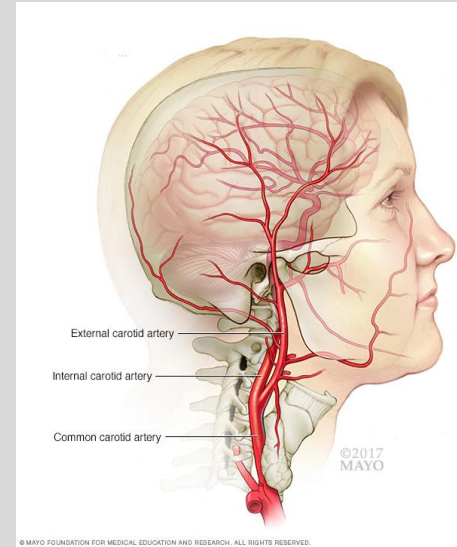


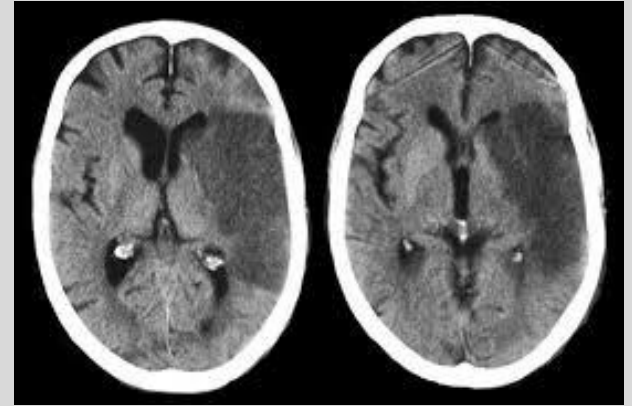
Image- mayo clinic
Umass Chan Medical School Stroke Stope

Middle Cerebral Artery Stroke

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- MCA is the largest branch of the ICA.
- **It is the most common site of ischemic stroke.**
- Contralateral weakness to face > arm > leg.
- “Look at your stroke” - gaze deviation towards the side of the lesion.

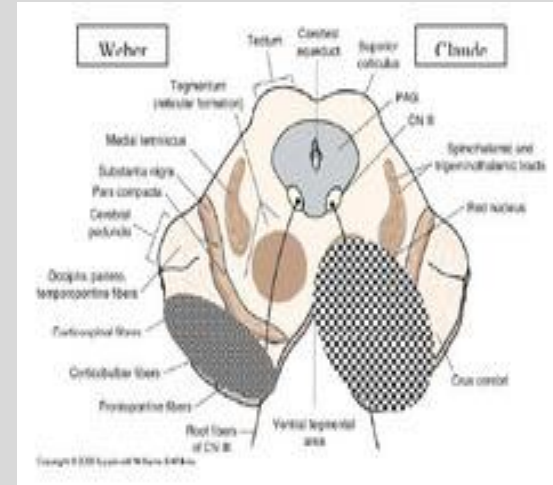


Brainstem Syndromes

- “Locked in”
- Wallenberg (lateral medullary)
- Medial medullary
- Weber Syndrome
- Foville Syndrome
- Benedikt Syndrome
- Claude Syndrome
- Nothnagel Syndrome

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Locked In Syndrome

- Severe neurologic condition with near-total body paralysis and preserved consciousness.
- Damage to the brainstem: ventral pons is most vulnerable area.
- Retained alertness with quadriplegia and anarthria
- Vertical eye movements are preserved.

Image- neurosciencefundamentals.unsw.wikispaces.net

<https://youtu.be/y-keNcCMo6Y> Uptodate.com January, 2025

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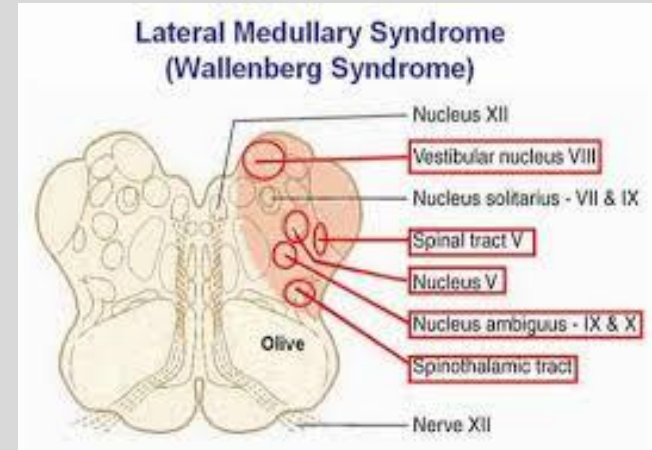


Lateral Medullary Syndrome

- Vertigo, nausea/vomiting, nystagmus
- Numbness- ipsilateral face and contralateral body
- Ipsilateral ataxia
- Ipsilateral paralysis of larynx, pharynx and loss of gag

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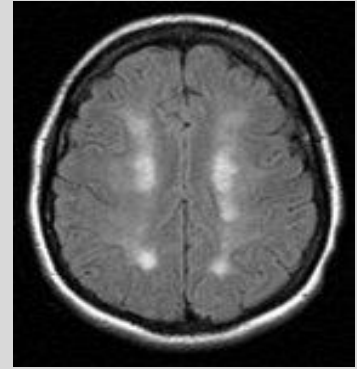


Man in a Barrel Syndrome

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- Anoxia to the cerebral border zones due to systemic hypoperfusion.
- Bilateral arm weakness while sparing the lower extremities.
- Damage to cortex that represents the arms



Lacunar infarcts

- Pure motor-post limb int. capsule
- Pure sensory- Ventral post lat nuc thalamus
- Mixed sensorimotor-thalamocapsular
- Dysarthria-clumsy hand syndrome-genu int cap, basis pontis
- Ataxia hemiparesis-basis pontis, internal capsule.

Case 3

- 57-year-old man
- 2-minute episode of left hand weakness. It was unusable during the episode. Walking and talking was normal throughout the episode.
- When he arrived at ED- he was completely normal. CT normal.

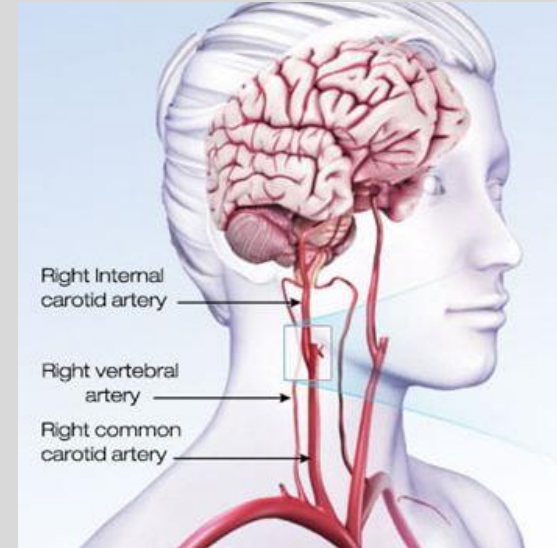
TIA

- A transient ischemic attack is a temporary blockage of blood flow to the brain.
- The clot usually dissolves on its own or gets dislodged. Symptoms usually last less than 5 minutes, but can last up to 24 hours.
- A TIA can help prevent a stroke, as it is a warning.

American Stroke Association

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Clinical challenge of TIA

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- Stroke mimics: Migraine, epilepsy, multiple sclerosis, peripheral nerve entrapment, brain hemorrhage (i.e. subdural hematoma).
- Motor and speech symptoms have a higher likelihood of brain ischemia as the cause of the symptoms.
- About 20% of patients presenting with TIAs will have a stroke within the next 90 days. Highest risk is within the next 24 hours.

Stroke Treatments

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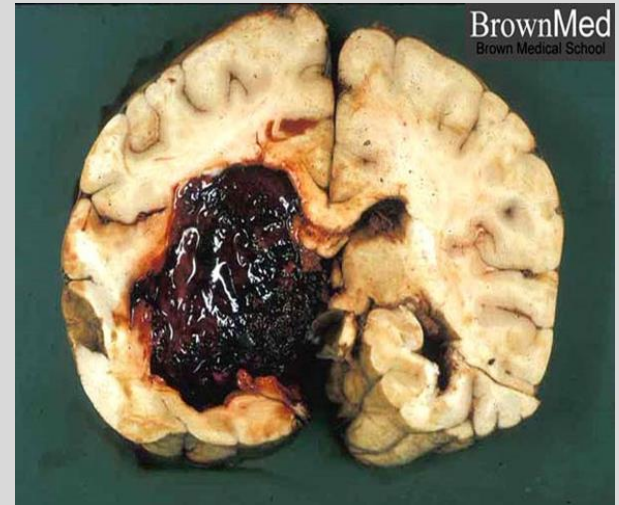
- **IV thrombolysis**
- **Mechanical thrombectomy**
- **Manage BP, glucose, temperature**
- Anticoagulation of atrial fibrillation
- Antiplatelet agents
- Statins for hyperlipidemia

Hemorrhages

- Epidural hematoma
- Subdural hematoma
- Intracerebral hemorrhage
- Subarachnoid hemorrhage

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Case 4

- 3/16/09 Natasha Richardson fell while skiing on a beginner hill and sustained a head injury. She refused medical help, but complained of a severe headache.
- She sustained an epidural hematoma and died 2 days later.



Question

- A fracture of which part of the skull is likely to be associated with a middle meningeal artery laceration and why?

Epidural hematoma

- Often traumatic with lens shaped appearance.
- Rupture of middle meningeal artery after skull fracture.
- A lucid interval before LOC (talk and die).

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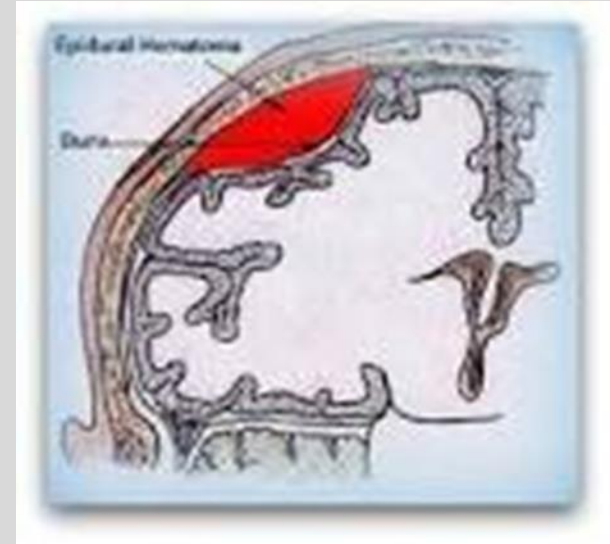


Image: Brain injury explanation. Continuum- Lifelong learning in Neurology. Cerebrovascular Disease. Volume 23, Number 1, February, 2017

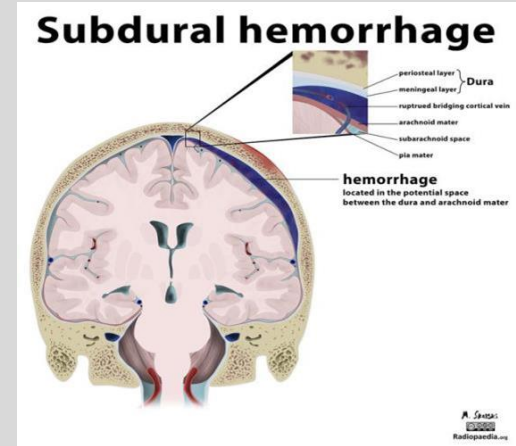
Subdural hematoma

- Usually due to trauma- MVAs, falls and assaults. A crescent shaped extra-axial lesion.
- Rupture of bridging veins
- Occurs between the dura and the arachnoid membranes.
- Trivial or minor head trauma in susceptible patients (cerebral atrophy).

Image- pixgalerie.com/mycleveandclinic.org
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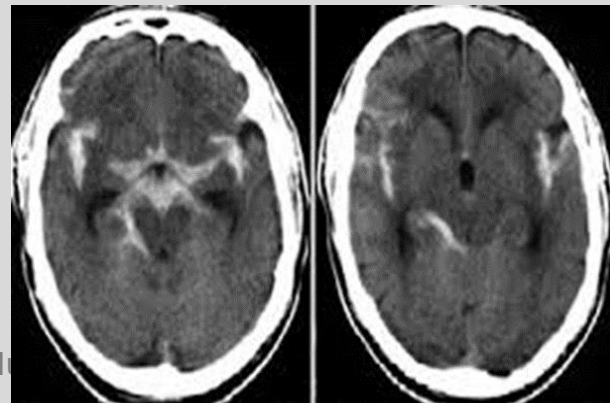


Subarachnoid Hemorrhage

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- Aneurysmal versus nonaneurysmal SAH
- Perimesencephalic is most common cause of **nonaneurysmal**.
 - Reversible cerebral vasoconstriction syndrome
 - Posterior reversible encephalopathy syndrome
 - Infective endocarditis, cocaine use
 - Cortical vein thrombosis
 - Coagulopathies



Case 5 -dissection

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- Jane is a 40-year-old woman who presented with a new severe headache on the right side of her head with associated neck pain. It began yesterday.
- “The worst headache I can remember.”
- History of thyroid cancer treated with radiation therapy.



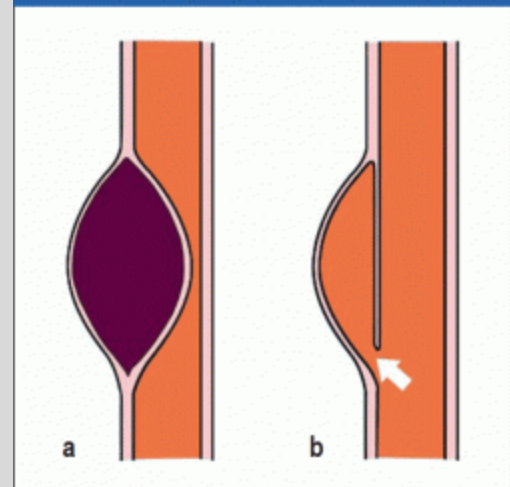
Arterial Dissection

- Blood collection and intramural hematoma formation in the arterial wall layers.
- May occur in the carotid artery, vertebral artery or both
- Most commonly involves the extracranial vessels
- Causes: Spontaneous or as a result of trauma.
- A common cause of stroke in young adults.

Causes of Cervical Dissection

- Connective tissue: Fibromuscular dysplasia, EDS IV, Polycystic Kidney
- Environmental factors: trauma (whiplash type)
- Use of oral contraceptives
- Migraine

FIGURE 1



Pathoanatomy of arterial dissection

a) Dissection of the internal carotid artery or vertebral artery with a hematoma in the vessel wall due to rupture of the vasa vasorum. The intramural hematoma can progress toward the lumen with secondary rupture of the intima.

b) Dissection of the aorta with rupture of the intima and formation of two lumina. Modified from (20).

Signs and Symptoms of Dissection

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- New-onset head pain (ICA)
- Neck pain (vertebral artery)
- Ipsilateral Horner syndrome
- Cranial and cervical neuropathies
- Pulsatile tinnitus
- TIA or stroke may result



Horner Syndrome

- Miosis
- Ptosis
- Without anhidrosis
- Results from distention of the 3rd-order sympathetic fibers that span the the external surface of an enlarged and dissected ICA.

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Case 6

It is November 2012, several days after hurricane Sandy.

A 49-year-old woman is brought into the office by her sister. She is in distress.

She has a severe headache and is having difficulty making sense.
On physical exam

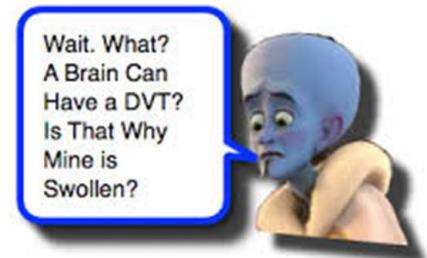
Unable to have a conversation, “felt like she was dying”

Visibly uncomfortable.

No focal neurological abnormalities.

Question

- What is the most common presenting symptom of a cerebral venous infarction?
- What diagnostic test is required to diagnose it?

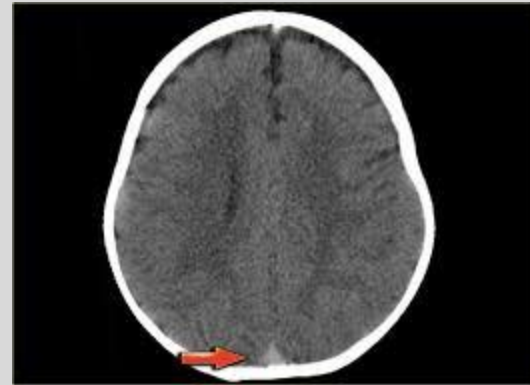


Venous Thrombosis

- Rates are higher than previously thought.
- Rates among women of reproductive age are consistently 3 times higher than their male counterparts.
- Associated conditions: genetic prothrombotic diseases, antiphospholipid syndrome, cancer, oral contraceptive use, infections (COVID-19)

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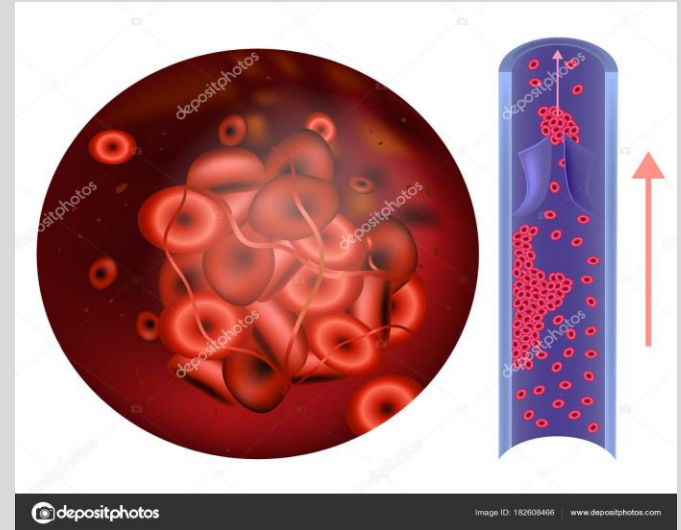


Prothrombotic conditions

- Factor V Leiden mutations
- Prothrombin G20210A
- Protein S deficiency
- Protein C deficiency
- Antithrombin deficiencies
- Antiphospholipid syndrome

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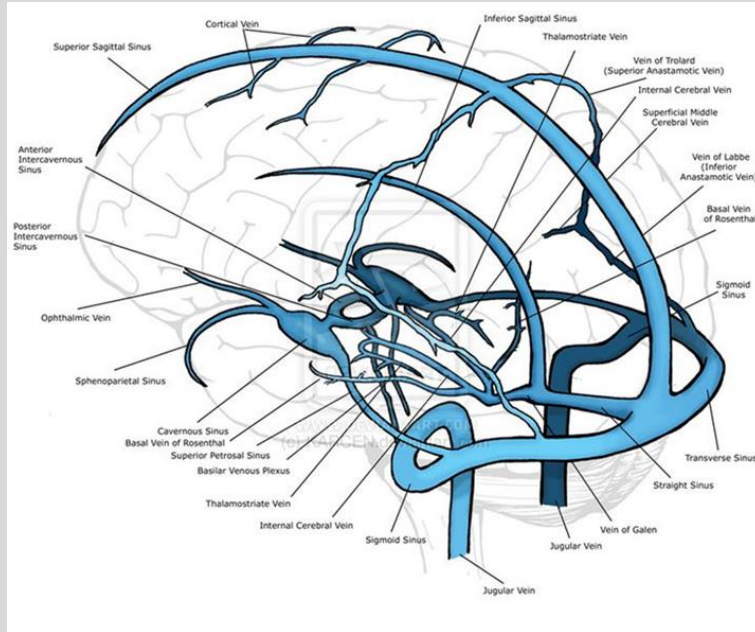
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- Cerebral Venous System

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Pinterest.com

Clinical Syndromes of CVT

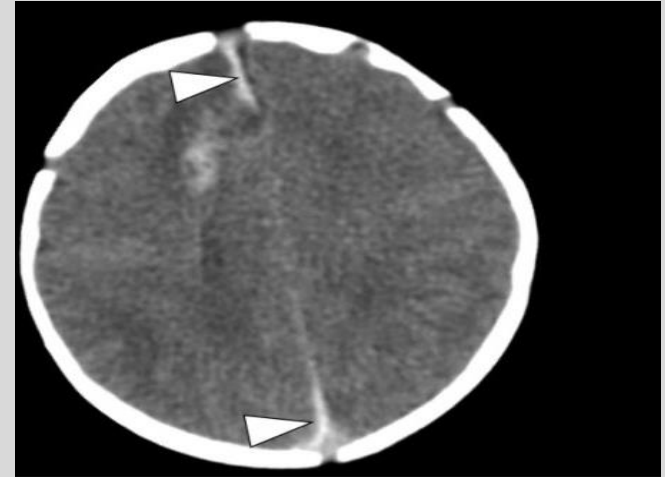
- Signs and symptoms are diverse. Presentation can be divided into four syndromes:
- 1) Headache (90%) or increased intracranial pressure
- 2) Focal neurologic presentations/ seizures
- 3) Subacute encephalopathy
- 4) Cavernous sinus syndrome with multiple cranial neuropathies.

Diagnosis of CVT

- Requires a high degree of suspicion. Cerebral angiogram is the gold standard, but noninvasive imaging is favored. CT and MR venography or MRI with contrast.
- **Hemorrhagic conversion seen in greater than 1/3 of patients (often parasagittal distribution).**

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Primary Stroke Prevention

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- Life's Essential 8
 - Healthy diet
 - Physical activity
 - Healthy weight
 - Healthy sleep
 - Avoid tobacco products
 - Healthy levels of blood lipids
 - Healthy levels of blood glucose
 - Blood pressure management

2024 Guidelines for the Primary Prevention of Stroke

Summary Slide

- Summary Point 1: Localize brain lesions based on patient's presentation
- Summary Point 2 : Remember that there are venous strokes in addition to arterial strokes.
- Core References:
 - Continuum- Lifelong learning in Neurology. Volume 29, Number 2. April 2023
 - Continuum lifelong learning in Neurology. Cerebrovascular Diseases. April 2020, Vol. 26 No 2
 - Uptodate.com

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>